Data Communication

Topic No.	Topic Name
Topic 1	Introduction of Data Communication
Topic 2	Components of Data Communication
Topic 3	Types of Data Communication

Data Communication

- Data communication is the process of transferring data or information between two or more devices through a communication medium.
- It plays an important role in modern computer networks and telecommunication systems. Through data communication, devices can exchange information efficiently and reliably.
- It supports many digital services such as internet communication, data sharing, and online applications.

KEY	For effective data communication, the data should be accurate, timely, and understandable to the receiving device.

Components of Data Communication

- A data communication system consists of the following components:

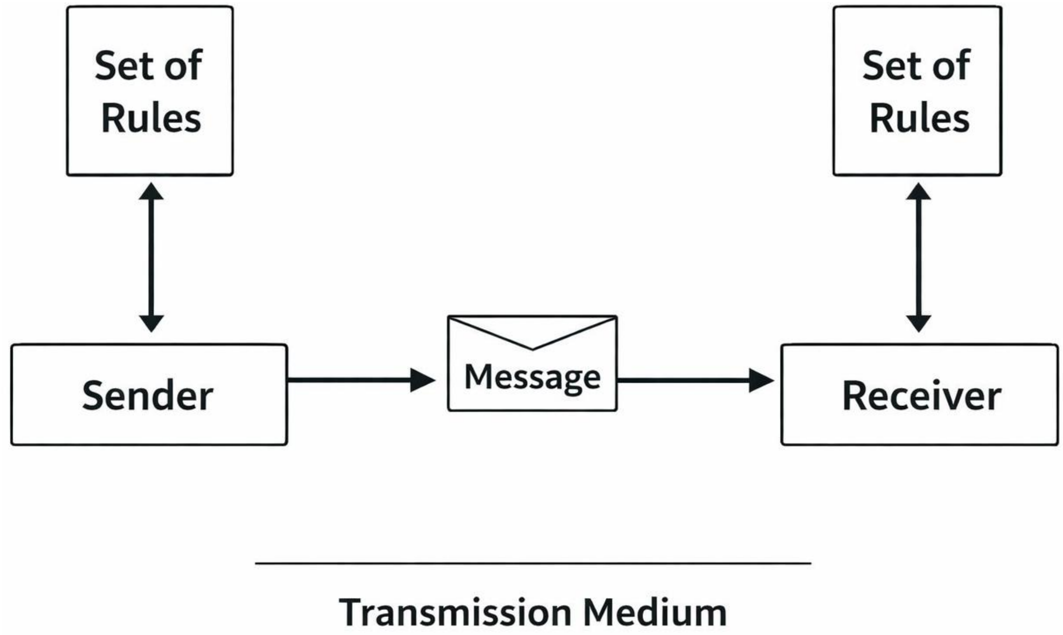

Figure 1: The 5 components of a data communication system

- **Message:** A message is the information or data that needs to be transmitted from one device to another. The message can be in different forms such as text, numbers, images, audio, or video.
- **Sender:** The sender is the device that transmits the data or message. It is responsible for initiating the communication and sending the information to the destination.
- **Receiver:** The receiver is the device that receives the transmitted data from the sender and processes the information.
- **Transmission Medium (Communication Channel):** The transmission medium is the path through which the data travels from sender to receiver. It can be a physical medium or a wireless communication channel that carries the signals between devices.
- **Protocol (Set of Rules):** A protocol is a set of rules and standards that govern data communication. Protocols define how data is formatted, transmitted, received, and interpreted between devices to ensure proper communication.

Types of Data Communication

Data communication can be classified into three types based on the direction of data flow:

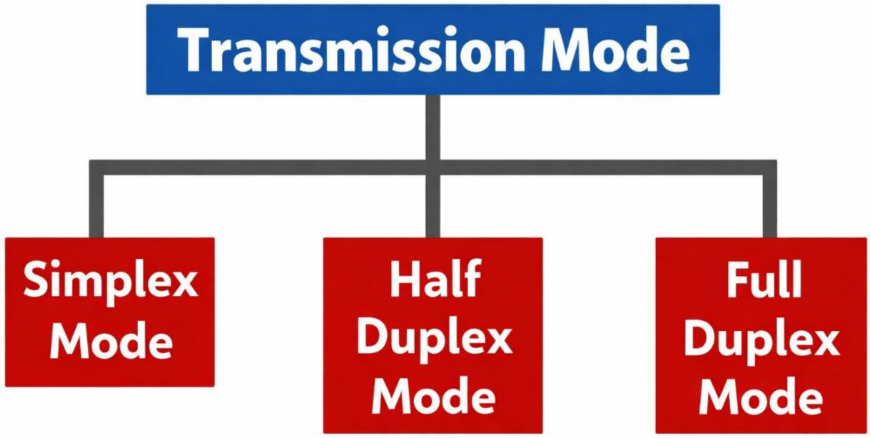

Figure 2: Three types of data communication

- **Simplex Communication:** Simplex communication is one-way communication where data is transmitted in only one direction. In this type of communication, one device acts as the sender and the other acts only as the receiver.
- **Half-Duplex Communication:** Half-duplex communication is two-way communication, but data cannot be transmitted in both directions at the same time. Devices can send and receive data alternately.
- **Full-Duplex Communication:** Full-duplex communication is two-way communication where data can be transmitted and received at the same time. Both devices can communicate simultaneously without interruption.

Simplex	One direction only | e.g. TV broadcast, keyboard to computer
Half-Duplex	Both directions but not at same time | e.g. walkie-talkie
Full-Duplex	Both directions simultaneously | e.g. phone call, video call